SRAM LLC - AI Adoption Playbook

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1. Executive Summary

The following playbook should allow SRAM to successfully leverage AI in its current form into the future. It outlines a pragmatic, phased approach focused on early wins, targeted near term pilots, and eventually what SRAM could look like as an AI-first company. Our recommendation is focused on how a tailored AI-driven business transformation agenda can strengthen SRAM's competitive moat, improve operating efficiency, and create new sources of revenue. SRAM's existing competitive advantage allows it to maintain market leadership and premium pricing; it's this same advantage that makes SRAM particularly well positioned for AI adoption. SRAM already boasts an integrated ecosystem of premium products. AI can extend that advantage by connecting currently fragmented data, improving decision-making, and turning SRAM's product ecosystem into a more intelligent platform over time.

Our proposed roadmap begins with broad AI democratization and foundational infrastructure. In the first 12 months, we recommend low-friction, high-adoption "early wins" to accelerate fluency and identify internal champions. In parallel, SRAM should address its single greatest barrier to enterprise-scale AI which are fragmented, or lack of, modern systems and data. Over a 24-month window, we recommend implementing a data backbone enabling SRAM to unify data into a usable enterprise asset.

Once those foundations are in place, we recommend prioritizing two high-value, but also achievable pilot opportunities. These are customer and dealer warranty support and demand forecasting. The first pilot is the most practical initial use case that can yield sizable gains simultaneously. Demand forecasting is simultaneously the most important near-term operational opportunity but is dependent on modern data infrastructure being in place. Beyond these pilots, we see an extended horizon of AI opportunity which can unlock new revenue and innovation through offerings such as AXS Intelligence, AI-assisted R&D, and OEM proposal automation. In that end state, AI not only improves SRAM's processes, but it enhances SRAM's products, customer relationships, and long-term strategic differentiation.

2. SRAM's Value Generation Engine

SRAM sells premium cycling components, component systems, and accessories primarily through three distribution channels. First off, SRAM provides specified parts to OEM bike manufacturers who complete bikes. Secondly, SRAM sells their components to distributors and dealers at retail bike shops. Lastly, SRAM markets directly to riders who can buy aftermarket upgrades and replacement parts directly through SRAM or through dealers. These products are horizontally diverse across the biking ecosystem, spanning multiple standalone brands including:

- Drivetrains and braking systems - SRAM (RED, Force, Rival, Apex, Eagle)
- Suspensions - RockShox
- Wheels and cockpit - Zipp
- Power meters - Quarq
- Cycling computers – Hammerhead (Karoo)
- Pedals – TIME
- Adjacencies – Velocio apparel, Ochain adjusting system

The breadth and depth of SRAM's distribution channels and product set create a multi-year customer journey, which is SRAM's mouse trap and value add to the biking ecosystem. SRAM's components can provide OEMs, retailers, and riders with a seamless biking experience from the build to the maintenance.

As a result, SRAM's product set introduces high switching costs for customers that value the integrated offering and provides leverage over buyers given the end-to-end reliance on SRAM's components in their manufacturing process. SRAM's durable advantage is its cross-brand integration breadth, which allows them to charge at a premium price

point. No competitor sells drivetrains, suspensions, wheels, computers, and other components as a connected ecosystem (even down to the customization offered via the AXS app). As a result, SRAM is uniquely positioned in the biking industry to capture outsized value should it adopt AI into its business model due to its interconnected product set.

3. Where AI Adds Additional Value

Based on our research, conversations, and expectations around SRAM's business, we view the AI value creation opportunity for SRAM as broad and deep. That said, to realize those benefits there is a specific sequencing we recommend SRAM follow to be successful.

Step 1: Early Wins (Month 0 – Month 12)

It is clear based on research and limited conversation with SRAM that internal processes are not necessarily "tech-forward". As such, there is meaningful lift from implement initial "early win" strategies to proliferate and democratize AI use within the organization. These strategies will not only help employee-level AI development but also enable the right people within the organization to be successful in pursuing the steps set out in the rest of our framework. These "early wins" primarily center around deploying individual AI tools for daily work.

A primary example would be providing coding assistants, GitHub Copilot and Claude Code for the software engineering team to enhance productivity and familiarize them with the capabilities of AI agents in developing technical infrastructure and software applications. Another would be promoting AI-powered notetaking, email drafting, and meeting summarization across the organization via Microsoft CoPilot. AI-integrated day to day tasks help to train SRAM's internal agent context window and can support future "digital twin" deployment for various roles within the organization especially within back-office functions such as finance and accounting. Lastly, pushing internal AI innovation competitions and forums, especially around AI agent development using CoPilot or Claude LLM backbones, are critical as well to galvanize group acceptance of these tools. This phase costs little, builds AI fluency, and creates internal champions before larger deployments. While the cost is low, the hard value generation benefit is similarly low as the intent is to cultivate AI adoption within personnel as opposed to replace personnel with AI.

Step 2: Data Foundations (Month 0 - Month 24)

Critical to the plan is making a deliberate investment in SRAM's data infrastructure to support AI adoption and usage at an enterprise-level. That involves modernizing the technology stack and doing it with an agentic-AI lens to provide a platform from which to extend AI adoption strategies. The value from these initiatives isn't even just limited to the significant potential uplift from AI but will help to provide SRAM with the tools to manage their business and product set centrally as opposed to branded silos.

SRAM operates without a centralized ERP, CRM, or data warehouse today. A unified data layer linking data on CRM, inventory, telemetry, and support data is necessary to provide the backbone on which to transform into an AI-first business. Our proposal is to construct an AWS-native data "lake house", given SRAM already uses AWS today, upon which modern BI, analytics, and AI/ML tools can sit. While not exactly where "AI adds value" it is a necessary initiative to unlock higher-level use cases for AI further down the road at SRAM. Please reference Appendix I for technology and data architecture diagram.

Step 3: Cost Reduction (Month 0 - Month 36)

This is where our proposal begins to add depth and bear fruit; after SRAM employees and systems are positioned to adopt AI into underlying processes as opposed to at a surface level. From a near term perspective, the most logical value creation opportunities for AI are in cost savings. We've identified 2 leading opportunities to address in this category:

1. Customer and Dealer Warranty Support (Month 0 – Month 12):
    a. Problem: SRAM's Trustpilot rating is 1.6 out of 5 when it comes to customer service and review often explicitly state intent to switch to Shimano after support failures. Public forums consistently report pairing issues, firmware failures, and shifting anomalies that remain unresolved due to

unsatisfactory customer support. This creates real customer retention risk, lost revenues, and increased costs to address failures which is driven by support quality, not product quality. This is a threat to SRAM's competitive moat, given failures in support for one product can reduce likelihood to retain customers through their cycling lifecycle.

b. AI Solution: SRAM maintains a large support knowledge base with manuals, compatibility guides, troubleshooting procedures, and warranty guidance. Today service support is not handled centrally, and support knowledge is not updated and disseminated throughout the organization in real time. New firmware or product updates are pushed without also addressing updated support knowledge base simultaneously. Amazon Bedrock and Kendra, powered by Claude, can act as agents interpreting product updates and pushing necessary changes to the support database in real time. Additionally, ~70% (per conversations with Clint) of inbound dealer and customer questions follow repeatable patterns that an AI agent can draft accurate responses to. Once piloted, the long-term solution could involve a fully automated support function that learns from prior interactions and delivers quicker and more effective customer service experience. Additionally, SRAM's inbound warranty claims deal with similar flows and repeatable questions; AI agents trained on the centralized dataset across products as well as warranty terms can more quickly and accurately assess warranty claims reducing fraud or frivolous claims that drag on SRAM's cost base.

2. Demand forecasting and supply chain optimization (Month 24 – Month 36):

a. Problem: Based on our conversations, SRAM views this as the most actionable and near-term value additive initiative. SRAM effectively has no demand forecasting engine in place today and as such is already high priority within the organization. For any manufacturer, stockout, markdown losses, or overproduction are hallmarks of inefficiency and among the largest "avoidable" costs within an organization. Better syncing production with the sales cycle can yield massive value gains without impacting the product or customer experience.

b. Solution: With the benefit of a formalized data lakehouse, a forecasting system built on Amazon SageMaker AI analytics tool, using historical order data, dealer sell-through, and seasonal patterns, could reduce stockout and markdown losses. SRAM already runs on AWS, so keeping the ML platform on the same cloud avoids the operational overhead and vendor fragmentation of introducing a second provider. ML models largely perform analysis autonomously and speak to existing inventory, production schedules, and manufacturing site statuses given data is embedded within the data lakehouse. This allows agents to do pre-planning of proposed response and turns the human into a true decision maker, freeing up SRAM resources or reducing headcount.

Step 4: - Revenue Uplift and Enhanced Innovation via AI-First Culture (Month 36 – Month 60 and onward)

Revenue-driven value creation initiatives often fall more towards the high-risk, high-reward end of the spectrum in terms of potential investments. This is no different for SRAM, but the opportunity set that AI presents is deep and highly value additive. The focus of our proposal is on taking the early wins and building critical infrastructure to support being an AI first enterprise by 2031. As a result, there are various potential opportunities for AI adoption with likely outsized returns, but we view these as long-term innovations in the project plan that will be explored, prioritized, and ultimately decisioned in the back half of our playbook window.

1. AXS Intelligence Platform:

a. Problem/Opportunity: SRAM's existing integrated cycling technology product set with AXS drivetrains, RockShox suspension, Quarq power meters, and Hammerhead computers, already represents its competitive advantage relative to peers. The data collected by these various components has historically been siloed and available to users via tailored media such as the AXS app that only has specific use cases for equipment customization.

b. Solution: By consolidating telemetry data from these products on an individual component level, SRAM can significantly widen its competitive moat by introducing an AI-powered riding experience. Not only can consolidated data be generated into user-friendly dashboards showing complete system data (e.g. heart rate, power output, gearing patterns, suspension behavior) using

tools such as SageMaker, but also agentic AI and ML applications via Amazon Bedrock can be designed to provide real time suggestions and even bike adjustments for riders. SRAM's product would branch out from basic hardware to truly intelligent components that continuously improve with usage. These enhanced benefits can be pushed to the AXS app via a new "AXS Intelligence" brand and unlock new revenue streams via premium tiered-subscriptions that provide more informative dashboards, AI-generated cycling advice, and real-time component self-optimization. SRAM can accelerate and widen its competitive advantage by further locking riders and OEMs into its product ecosystem due to not only an integration advantage, but a data advantage.

2. Generative design and R&D acceleration:

a. Problem/Opportunity: SRAM's R&D and new product development product are the essential lifeblood to maintaining its premium pricing and strong market position. Currently this process is entirely human driven and only using technology as tools as opposed to assistant. As a result, R&D process can be very disjointed and extended based on human creativity. Additionally, SRAM still physically tests all iterations in the lab. This can serve to continuously extend lead times in development as new iterations take time to set up and test in the physical world. All of this leads to a slower development cycle which creates risk that SRAM cannot stay ahead of competitors or puts more pressure on new launches to be even more successful.

b. Solution: AI-assisted design and R&D can massively enhance engineer productivity leading to higher throughput and ability to test potential innovations in rapid succession. For instance, digital twin testing (simulating wheel pressure tests or suspension stress rather than running physical tests for every iteration) can reduce lead item in testing multiple iterations of a new product. Additionally, field data collected as part of AXS Intelligence platform can train AI agents on actual telemetry and performance data of similar products to have it propose or enhance new designs. AI-assisted patent searches can help SRAM stay on top of competitor pipelines in real-time and even assist in patent generation for their own products, freeing up engineering talent to focus on innovative design as opposed to IP documentation.

3. OEM proposal automation:

a. Problem/Opportunity: Similar to R&D and design, SRAM's current proposal and specification package creation for bicycle manufacturers is overly manual. It requires significant man hours from cross functional teams including, design, engineering, sales, and product management. Speed to RFP and responsive to client feedback is key to increasing throughout and OEM satisfaction for SRAM.

b. Solution: Agentic AI solutions built into SRAM's new technology ecosystem like Salesforce Einstein/Agentforce can act in several ways to improve win rates with OEMs, reduce sales cycle timing, and improve client retention. First off, an AI agent familiar with the development, pitch, and sales process can construct and remotely manage the project plan in order to facilitate orderly responses to client requests. Additionally, it can be instructed to take over manual tasks within the process including documentation generation, learning deal to deal, the agent can inform stakeholders of OEM preferences and winning strategies to land RFPs going forward. Tools like Agentforce can also use configure, price, quote features to automate proposals and outcomes on more homogeneous, lower volume, qualified opportunities.

4. Compatible upgrade recommendations:

a. Problem/Opportunity: SRAM's current sales model, especially through the retail dealer and online DTC channel, is single product delivery focused. Said differently, when a rider comes to make a purchase or a dealer is selling SRAM's products to a rider, the marketing and sales model focuses on finding the specific component for their current needs. As such, product cross-sell without customer intent is limited. Additionally, sales follow-up is limited until the customer decides an upgrade or replacement is necessary.

b. Solution: ML / AI solutions like Amazon's SageMaker can create a compatibility rules engine paired with a recommendation model to lift rider basket value while building trust through accurate fit guidance. They can also model individualized recommendation reach outs to customers based on actual usage data tied to registered components on that customer's bike. Not only for predictive maintenance telling someone it might be time for a replacement, but also for compatible upgrades for existing components or ancillary components along with actual feedback on the performance gains from those upgrades. These same insights on paired products can be provided to retail dealers to generate tailored and increase turnover of SRAM inventory.

4. Implementation Strategy

With this scope in mind, we've endeavored to set out an implementation strategy to properly realize the benefits of the near-term project plan. We structured this in line with the high-level rollout strategy of first pursuing initial democratization and data infrastructure investment; both of which preceding initial key AI pilot initiatives. This targeted strategy in the next few years is designed to capitalize on clear, tangible, and highly actionable AI use cases like customer support and demand forecasting while the business gains familiarity with and implements modern tools as well as builds organizational trust to allow for long-term AI-driven initiatives to be achievable. To detail this project plan, we've set out core components in 3 main sequences: preliminary preparation, actionable pilots, and long-term innovations. Below we cover the first two specific action plans, whereas the third sequence is more of a picture of where SRAM can be when leading with AI first in the next 5 years and onward.

Preliminary Preparation – Key Integration Components Matrix:

Integration Attribute	Early Wins (Month 0 – Month 12)	Data Infrastructure (Month 0 – Month 24)
Key Initiatives	• Microsoft CoPilot deployed across all functions • Provide Github Copilot and Claude Code licenses to all software development staff • Facilitate weekly AI Forums SMEs teach new use cases to the group as well as open discussion on how people are leveraging tools	• Month 0-6: ERP and CRM implementation vendor contracting; AWS data lakehouse architecture finalized • Month 6-18: SAP implement and go-live for SRAM core brands; Salesforce CRM live for dealer and warranty management; data pipelines from POS and support systems • Month 12-24: AXS, Hammerhead, Quarq, and other telemetry data integrated into data lakehouse; Power BI and Databricks deployed; data quality validated and specifically tested for demand forecasting use case
Existing Integrations	• No mandated change to current workflows • AI tools leveraged on individual level to develop new solutions to workflows; tools assist within current processes	• Integrating with pre-existing data-generating products; AXS, Hammerhead, etc. are already feeding data live • All financial, inventory management, HR, procurement, and other back-office functions realign processes to new SAP ERP • All sales staff realign processes to new Salesforce CRM platform • Data engineering team tasked with data lakehouse architecture, construction, and data taxonomy to

		ingest and compile disparate data into usable format
New Implementations	• None needed • Microsoft Suite and software development environments are already equipped or can easily install AI tools	• Core purpose of initiative • Significant new technology implementations via SAP, Salesforce, and AWS data warehousing
SMART Goals	• By Month 6, >75% of software engineers use AI-tools as measured by usage dashboards • By Month 9, >60% of other functional staff using CoPilot as measured by M365 analytics • By Month 12, >5 internal AI champions identified within relevant departments and assigned to pilot teams in future initatives	• By Month 15, Salesforce CRM is live for dealer and warranty management across all SRAM product lines • By Month 18, SAP ERP is live for SRAM core brands with >80% of order and inventory data flowing into the lakehouse • By Month 24, AXS and Hammerhead telemetry integration achieves >90% of active device data and lake house data quality scores >85% on field required for demand forecasting
Organizational enablers	• No new hiring or data required • Program management support • For executive buy-in CTO sponsor engineering deployment and COO sponsors business rollout	• Requires standing up a data engineering function within the CTO organization with 3-5 FTEs • Requires one senior data architect and outsourced system integration services from Accenture for SAP and Salesforce implementation • VP-level executive tasked with ownership of control tower to ensure executive oversight and enforcement
Financial implications	• Costs: ~$300k annually for relevant AI tool licensing • Benefits: Cost savings of ~$1mm in FTE productivity on run-rate basis based on overall productivity enhancement of estimate 10%	• Costs: ~$10mm project implementation cost and ongoing operating costs of ~1.5mm annually due to expense source systems and integration teams • Benefits: Minimal standalone incremental benefit; returns come in the form of enabling the deployment of other accretive initiatives that require data infrastructure

Actionable Pilots – Key Integration Components Matrix:

Please reference Appendix II for detailed Gantt chart outlining pilot and implementation programs.

Integration Attribute	Customer and Dealer Warranty Support (Month 0 – Month 12)	Demand forecasting and supply chain optimization (Month 24 – Month 36)
Key Initiatives	• Pilot design focused on AXS and Hammerhead support • Utilize AI-enabled search tools like Amazon Kendra to provide quicker customer support resources for human teams • Eventually leverage AI agents via Amazon Bedrock to draft ticket responses, review warranty claims, and interact with external parties • Early pilot initiative can be isolated from data lakehouse	• Following audit and validation of data readiness from data lakehouse, implement AI / ML driven demand forecasting model to improve throughput, manufacturing planning, and reduce inventory costs • Utilize Amazon SageMaker AI analytics tool to build SKU forecasting models • Run pilot concurrent with existing forecasting process to monitor AI performance relative to historical practice • Eventually full model recommendations with human review and override
Existing Integrations	• Amazon AI agents and tools plugged into existing workflows via Zendesk, so existing ticketing structure and support teams remain in take • Amazon Kendra integrates with SRAM's support knowledge base in real time including manuals and guides	• Amazon SageMaker model linked to AWS data lakehouse built in early wins phase • Data sources include new SAP ERP for inventory and order history and Salesforce CRM for sales data • Production planners retain full authority and pilot runs concurrent with existing flows; eventually model recommendations replace existing flows with planner override
New Implementations	• Amazon Bedrock (Claude) drafts responses from Kendra-retrieved content for human review and approval within Zendesk • Warranty claim AI review integrates similar with existing dealer records, warranty data, and existing workflows with human decisioning	• Amazon SageMaker ML demand forecasting model built ground up and trained on unified historical order, inventory, and sales data • Power BI supply chain planning dashboard for ease of review on model outputs and making summarized SKU-level recommendations • Quarterly model retraining cadence established to continuously improve results
SMART Goals	• During Pilot through Day 120 and Month 15: ○ 40% reduction in average	• By Month 27 back-test validation of model produces <15% error and <20% on live data

	time to first response on AXS and Hammerhead ticket pilots ○ 15% improvement in tickets resolved without escalation ○ 70% of AI-drafted responses approved by support with minimal edits ○ 25% more tickets handled per agent per day ○ 90% agreement between AI warranty agent recommendation and human decision	• By Month 36: ○ 30% reduction in stockout rate ○ 25% reduction in markdown and write-down rate ○ 15% improvement in inventory turns ○ >60% of model recommendations acted upon by planners without override ○ >$1.6mm in annualized avoidable inventory loss recovered
Organizational enablers	• Dedicated product owner • One integration engineer to handle Zendesk and AWS tooling • One support team champion to manage adoption within customer service function • Named C-suite, likely COO, sponsor with accountability of monitoring and pushing pilot	• One ML / data science lead for model development • Two supply chain analysts redeployed from existing team with AI workflow retraining to own the planner review process • Data engineer from the data foundations team retained to own SageMaker data pipelines and ongoing data quality monitoring • COO and CFO named as co-owners with monthly steering committee review • Override documentation policy is mandatory for all planners and non-negotiable as it directly feeds model improvement
Financial implications	• Costs: ~$500k set up for Zendesk/AWS tooling and staff; ~$300k annually in Zendesk and AWS fees • Benefits: Cost savings of ~$400k annually in warranty claim accuracy improvement. Cost reduction via efficiency gains in support department and likely savings in headcount reduction. Unquantified revenue uplift due to improved customer retention.	• Costs: ~$500k setup (SageMaker development, data science lead ramp, analyst retraining); ~$700k annually in ongoing operations (data science lead, SageMaker compute, data engineer time) • Benefits: ~$2.5mm stockout loss recovery, ~$1.5mm markdown reduction, ~$0.8mm inventory carrying cost savings, plus a one-time working capital release from inventory reduction

5. Extended Adoption Horizon – Innovation Uplift and AI-First Culture

Over an extended horizon out until 2031 and beyond, our proposal identifies various avenues to see true transformative uplift at SRAM. These solutions are primarily focused on revenue uplift, accelerated product development, and ultimately both leveraging and maintaining SRAM's competitive advantage and premium price point. As such, these initiatives, while lofty, are worth reassessing following the results of the initial early win plans and pilot programs. We refrain from specifying prioritization and associated timelines for these initiatives on that basis; depending on where SRAM is on its AI journey initially 36 months in and where their business is positioned, these opportunities may be ranked differently depending on strategic importance and potential returns. As such, we

view these AI applications as more of what SRAM could look like as an AI-first company by the end of our horizon but have offered specific guidance on how they might enact these strategies below. As stated previously, it's SRAMs unique market position today that lends itself to being able to extract outsized returns from developing a coherent, targeted, and ambitious AI strategy. This is what the fruit of that labor could look like.

Strategy	Key AI applications	Timeline	Investment / Returns	Leadership / Team / Processes needed
AXS Intelligence	SageMaker component health ML Amazon Bedrock agentic recommendations	Month 36 - Month 54	Investment: $5mm build and $2mm annual operating costs Returns: ~$20mm ARR base case via AXS Premium subscriptions Tiers: AXS Premium - $79 per year; AXS Performance - $199 per year	Cross-Brand PM staffing C-suite AI Sponsor, likely CEO given scale AXS Intelligence brand launched in AXS app
Generative design and R&D acceleration	Autodesk Generative Design / topology optimization Digital twin testing (replace physical lab iterations) AI patent search (PatSnap) AXS telemetry trains design AI agents	Month 42 - Month 60	Investment: $4mm tooling and engineering time Returns: $2mm per year in physical prototypes avoided; $2mm per year time-to-market compression; $2mm per year premium pricing uplift	CTO / VP Engineering ownership R&D team AI workflow retraining Digital twin infrastructure build-out Requires AXS Intelligence platform telemetry as design training data
OEM proposal automation	Salesforce AgentForce with CPQ (configure, price, quote) AI proposal project management Automated documentation generation AI deal intelligence to improve OEM win rates	Month 36 - Month 48	Investment: $1mm (AgentForce and CPQ) Returns: $2.5mm annually from 1% OEM win rate improvement	VP Sales / CRO ownership Cross-functional team training (sales, design, engineering, product mgmt) AI agent onboarding for OEM sales team
Compatible	Amazon SageMaker compatibility rules	Month 42 - Month 60	Investment: $500k in tooling	Sales and marketing leadership

upgrade recommendations	engine + recommendation model Personalized outreach via AXS telemetry and registered component data Predictive maintenance alerts; dealer inventory optimization		Returns: $3mm annual cross-sell/upsell lift (3-5% of registered user base); $1mm annually from dealer inventory turnover improvement	Dealer network integration Requires AXS registered component data and Salesforce CRM AI model training on purchase history and AXS usage data

Appendix I

Source Systems →	"Data Lakehouse" →	Integrated Tools
Central ERP – SAP	AWS Native Platform	Power BI (Business Intelligence Layer)
Central CRM/Dealer & Warranty Management – Salesforce		Databricks (Analytics)
Product Data – AXS (telemetry, RockShox), Hammerhead (Karoo GPS), Quarq Power (Ride data)		*Amazon Bedrock / Kendra / Claude (Warranty AI Agent, AXS Intelligence, Generative Design & AI-Accelerated R&D)
		*Amazon SageMaker (Demand Forecasting, AXS Intelligence)
		*AgentForce (Salesforce CRM Agent)
		*PatSnap (AI Patent Search)
**AI / ML enabled tools and agents. End goal of playbook, but require modern data infrastructure*		

Appendix II

Customer and Dealer Warranty Support Pilot

Month 1–24 · All AI-drafted responses require human agent approval during pilot period · ◆ = Decision Gate

Active Pilot Work · Scale / Deployment · ◆ Decision Gate · Ongoing / Maintenance

Initiative / Task	Q1 M 1–3	Q2 M 4–6	Q3 M 7–9	Q4 M 10–12	Q5 M 13–15	Q6 M 16–18	Q7 M 19–21	Q8 M 22–24
PHASE 1 · SETUP & INTEGRATION Month 1–3								
Pilot Design + Baseline Metrics								
Kendra + Bedrock (AWS) Integration								
PHASE 2 · AXS DRIVETRAIN PILOT Month 4–6								
AXS Drivetrain Ticket Rollout [100% human review]								
◆ *Day-90 Draft Acceptance Checkpoint (G3 ≥50% gate)*	◆							
PHASE 3 · HAMMERHEAD EXPANSION + WARRANTY TRACK Month 7–12								
Hammerhead Device Ticket Expansion								
Warranty Claim AI Review Track [human final decision]								
◆ *Day-120 Formal Go / No-Go Decision*		◆						
PHASE 4 · SCALE TO FULL PRODUCT CATALOG Month 13–24								
Full Product Catalog Rollout								
Semi-Autonomous Response Tier [low-risk categories]								

Demand Forecasting and Supply Chain Optimization Pilot

Month 22–48 · Requires Data Lakehouse with ≥85% data quality before launch · ◆ = Decision Gate

Active Pilot Work · Scale / Deployment · ◆ Decision Gate · Ongoing / Maintenance

Initiative / Task	P1 M 22–24	P2 M 25–27	P3 M 28–30	P4 M 31–33	P5 M 34–36	P6 M 37–39	P7 M 40–42	P8 M 43–45	P9 M 46–48
PHASE 0 · DATA READINESS GATE Month 22–24									
Data Readiness Audit [≥85% completeness on pilot SKUs]									
◆ *Data Quality Gate — hard go / no-go before model training*	◆								
PHASE 1 · MODEL DEVELOPMENT & BACK-TEST Month 25–30									
SageMaker Model Development [top-50 SKUs by revenue]									
◆ *Back-Test Validation Gate (MAPE ≤15% required to proceed)*		◆							
PHASE 2 · SHADOW MODE PILOT Month 28–33									
Shadow Mode — Model Runs Parallel, Planners Observe Only									
◆ *Shadow Checkpoint (MAPE ≤10% live; planner adoption signal)*			◆						
PHASE 3 · ACTIVE PILOT — PLANNERS ACT ON MODEL OUTPUT Month 31–36									
Active Pilot — Planners Act on Recommendations [overrides logged]									
◆ *Full Pilot Evaluation + Scale Decision*					◆				
PHASE 4 · FULL SCALE — ALL SKUs + BRANDS Month 37+									
Full SKU Catalog + Multi-Brand Deployment									
Quarterly Model Retraining Cadence									